Benchmark for Crack Re-Identification in Ticket Management Systems

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***Abstract*—In a ticket-based maintenance system, each new photograph of a crack should be assigned to the correct ticket, but if the same crack is photographed from a different angle it may be treated as a new defect, breaking its historical record and leading to redundant inspection work. We introduce CRACKID, a dataset of 140 photographs from 17 walls containing 212 distinct crack identities, and use it to assess ten zero-shot matchers under three scenarios: retrieval, thresholded pairwise decisions, and open-set identification with rejection. Ranking quality varies considerably (Rank-1 0.352–0.654), yet validation-calibrated pairwise decisions stay close to the best constant baseline (F_1 0.293–0.351 vs. 0.305). Context and crack-erasure experiments show that wall-context cues matter more than crack appearance. A geometric reference that aligns crack-free wall texture and compares projected masks achieves a detection-and-identification rate of 0.768 at a false-alarm rate of 0.1 on the 48% of pairs that can be aligned, compared with 0.429 for the best crop-based method. These point estimates are from a small wall-clustered sample and cover only single-visit correspondence. They indicate that good retrieval alone does not guarantee reliable crack association, and that local visual context and geometric alignment are both needed. CRACKID provides a foundation for developing more reliable visual association methods, with cross-visit evaluation and ticket-level duplicate reduction as future work.**

***Index Terms*—Benchmark, Building Maintenance, Crack Re-Identification, Geometric Alignment, Homography**

I. INTRODUCTION

Ticket management systems record reports of physical defects so that they can be inspected, prioritized, repaired, and followed up. Ideally, each ticket holds the history of a single incident. In practice, different users may photograph the same crack more than once, producing duplicate tickets that fragment this history and cause redundant assessment work. Every new observation must therefore either be linked to an existing ticket or open a new ticket for damage that has not yet been recorded.

This is a visual re-identification (re-ID) problem in which (i) the same physical crack must be recognized despite variations in viewpoint, camera distance, field of view, illumination, and partial visibility; (ii) a crack may itself change by extending, widening, or branching and must still be matched to its earlier observations, while genuinely new damage must be distinguished from growth of known damage. Errors in either direction lead to duplicate tickets or misattributed history. Existing work on crack detection and segmentation focuses on single images. Datasets such as CrackSeg9k [1] and OmniCrack30k [2] contain no cross-observation identities, so observations cannot be associated with tickets. It remains unclear whether techniques developed for person and object re-ID [3], [4] generalize to cracks, since different defects may look similar, while the same defect can vary considerably across observations.

This work introduces CRACKID to evaluate whether photographs depict the same physical wall crack: 140 photographs of 17 walls captured during a single visit. Real photographs capture the acquisition changes that occur within a visit, whereas appearance and growth changes are probed only with synthetic images. Cross-visit evaluation on real photographs and ticket-level duplicate reduction are left for future work. All accuracy figures are point estimates from a small sample grouped by wall.

We assess ten zero-shot matchers using CRACKID. The protocol separates candidate image retrieval from the decision of whether two observations depict the same defect, while incorporating open-set identification with rejection. The system accepts the top-ranked match only if its score exceeds a predefined threshold; otherwise, it rejects the query and treats it as a new ticket. This rejection mechanism is necessary because a high ranking alone does not provide sufficient evidence to link a report to an existing ticket.

Three research questions guide the study:

- **RQ1.** Can the evaluated methods associate observations of the same defect under varying image acquisition conditions?
- **RQ2.** How do wall context and geometric alignment contribute to defect re-identification?
- **RQ3.** How do two selected approaches perform under synthetic changes in appearance and crack growth?

RQ1 and RQ2 are answered with real single-visit photographs; RQ3 is an exploratory study on synthetic data.

The contributions are as follows: (i) the cross-observation identities of CRACKID, together with their annotation provenance (Section III-B); (ii) a benchmark covering retrieval, same-defect decisions, and open-set identification, which addresses RQ1 (Sections IV, VI-B, and VII); (iii) context controls, a geometric reference, and a coverage analysis, which address RQ2 (Sections III, VI-C, and VII-B); and (iv) a synthetic revisit probe, which addresses RQ3 (Section VII-C).

II. Related Work

Most crack-analysis research focuses on detecting or segmenting cracks in individual images [5]–[7]. Consolidated datasets [1], [2] retain this image-level formulation and do not provide identity annotations across observations. Although multi-view pipelines establish geometric correspondences to fuse images acquired during a single inspection into surface models or aggregated segmentations [8], they do not explicitly represent persistent crack identities across observations. Structural-health monitoring tracks cracks over time using fixed cameras and designated locations [9]–[11] or guided setups [12], where crack identities are assigned at construction time; unlike our open-set association task, they track known cracks.

The scenario involves object re-ID [4] with open-set identification [13]: SIFT [14] and ORB [15], learned correspondences such as SuperGlue [16] and LoFTR [17], and region embeddings [3], [18]–[21] are all assessed in a zero-shot manner without CrackID fine-tuning (the YOLOv8 row: layer-9 `yolov8n.pt` features [22]). LightGlue [23] and NetVLAD [24] are mentioned as future retrieval baselines. Vehicle-damage re-ID across claims [25] follows the same open-set protocol as ours; related work focuses on detection and severity, not on association [26]–[28]. The approach of aligning first and then reasoning about presence or absence constitutes change detection [29], [30] using planar registration [31]–[33] and an ECC fallback [34]; our key point is that the alignment and the matching targets must be disjoint, with the homography being derived from the wall when the crack is identified.

III. Data Acquisition

A. Acquisition and composition

The CrackID dataset consists of 140 photographs of 17 exterior walls taken at a single site over four half-days using two consumer handheld devices (Mendeley Data [35]; Fig. 1). The walls are either stippled render (textured) or smooth painted plaster, and the only features that are visible are cracks. The dataset is divided such that six walls are used for validation (walls 01 to 06; 66 photos, 66 identities) and eleven walls for testing (walls 07 to 17; 74 photos, 146 identities). Only those identities that appear in at least two photographs are included in the answerable queries (18 validation, 19 test). We added walls 18 and 19 to the dataset afterward and therefore exclude them from all results. Each wall was photographed during a continuous walk-around (median 3 s between frames, max 81 s); the data include within-visit viewpoint and scale correspondences, providing a more favorable scenario than multi-visit matching, not a formal upper bound. Cross-wall search is disallowed, so the wall is given for free. Counts for the frozen snapshot: validation has 6 walls / 66 photographs / 66 identities (18 matchable, 439 known queries, 74 open-set); test has 11 walls / 74 photographs / 146 identities (19 matchable, 280 known queries, 258 open-set); positive query–gallery pairs: 26,012 validation / 5,722 test; valid query–gallery pairs: 89,118 / 31,822. Of 301 same-wall test photograph pairs, 255 have valid cells and 97 register successfully. Of 31,822 valid test query–gallery pairs, 15,328 fall inside registered pairs (48%).

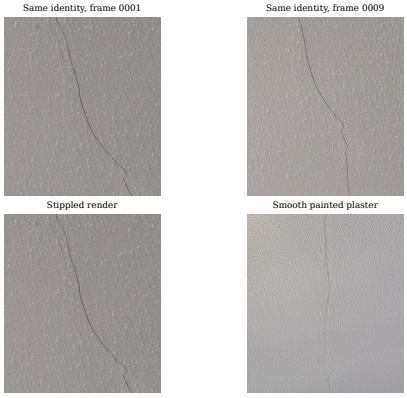


Fig. 1. CrackID examples. Top: one identity across non-adjacent frames. Bottom: stippled render versus smooth plaster. Labels via homography propagation, unaudited.

B. Annotation, and why it needs auditing

The ORB homography method carries every single click from frame to frame and then combines the fragments to form physical-crack identities; the merging process is the greatest risk (since an incorrect merge can turn a hard negative into a false positive). In the frozen snapshot, 95 of the 212 identities (45%; 70 of 146, 48% in the test case) result from merges, with median bridged gaps of 228 px (test) and 286 px (validation). One annotator examined all the clicks and sorted the unclaimed points (107 of 1,023 were classified as no crack); agreement was not assessed. Although single-annotator triage reduces the merge risk, it does not eliminate it. The live dataset must not be combined with the frozen figures without first re-running the benchmark.

IV. Benchmark Protocol

The benchmark assesses visual association by determining, for a new crack observation, whether it matches one of the recorded defects; since annotated identities serve as a stand-in for permanent defect records, the scores indicate the accuracy of the association rather than the reduction of duplicates at the ticket level.

A. Task formulation and candidate records

There is one physical wall; an identity connects the different observations of the same crack (Section III-B). A query refers to one instance of a crack; its gallery includes other instances taken from photos of the same known wall. Cross-wall search is not allowed. A query is called known if its gallery contains the identity, otherwise it is unknown (meaning it is a genuinely new defect). In closed-set retrieval it is assumed that the answer will always be in the gallery and the candidates are only ranked. Open-set identification, on the other hand, involves modelling the association of tickets: the system has

to either link the query to the best existing ticket or reject it and create a new ticket if no gallery entry scores high enough. Each method evaluates query–gallery pairs for the purposes of ranking, makes thresholded decisions and carries out identification with rejection. Pairs are the unit for pairwise decisions; queries are the unit for retrieval and for open-set decisions; image pairs are the unit for registration and assignment. The Hungarian one-to-one constraint is applied only for each image pair.

B. Test conditions, decisions, and metrics

The primary test involves actual walk-arounds for each wall, assuming that the morphology remains unaltered; the only variations considered are viewpoint, distance, scale, lighting, and visibility, along with same-wall distractors and unknown queries. The validation and test walls are entirely separate. All the methods are used exactly as they are on the shared masks without any tuning of CRACKID (see Sec. VI-A); the thresholds determined on the validation set are applied to the test set, considering only within-visit correspondence. A synthetic check is carried out to distinguish between photometric effects (condition B: blur/noise/contrast with unchanged structure) and growth effects (condition A: one tip extended) using ten photos taken from five walls with a single seed and a common warp (parameters available with the code), comparing OSNet with context to HYBRID-SKEL (τ=8.0 px), which is different from REG+CHAMFER: this is only a preliminary check.

Decisions: any pairs that exceed the validation-calibrated threshold are accepted; for open-set identification, only the highest-ranking gallery entry above the threshold is accepted for each query, and a query with nothing above threshold is rejected as a new ticket. For each image block, the Hungarian assignment is used with those below the threshold being dropped. Unscorable pairs are retained (they are predicted to be negative for F_1 and are given the last rank in retrieval); a registration failure is defined as the absence of evidence, not as evidence of a new defect. The metrics used are Rank-1 and mean Average Precision (mAP) for closed-set ranking; pairwise F_1 compared with the oracle F_1^* versus the validation $F_1^{@v}$; assignment-F_1^a equal to $2\mathrm{TP}/(2\mathrm{TP}+\mathrm{FP}+\mathrm{FN})$; and, for open-set, detection and identification rate (DIR) at false alarm rate (FAR) of 0.1 that is, the fraction of known queries correctly linked when at most 10% of unknown queries are wrongly linked together with coverage (the fraction that is scored). The references include random ranking and the best-constant F_1 $2p/(1+p)$ (0.305 at a prevalence of 0.180; 0.544 on the 0.48 geometric subset; 0.452 in the validation at 0.292); the full chance levels are given in Table I. Since there are 11 walls and 19 matchable identities, we provide the point estimates (Sec. VIII).

V. A GEOMETRIC REFERENCE BASELINE

The baseline tests whether crack locations on the wall drive the association: it aligns wall surfaces to a common frame and then compares the projected crack masks. It assumes a roughly planar wall and uses no CRACKID training. The shared masks (trained on public crack data only; no CRACKID wall in training) use instance extraction at 200 px and 5 px to keep inputs consistent across methods. Masks are taken as given, and their error is not measured.

A. Registration using wall texture

For I_a, I_b, a homography H ($x' \sim Hx$) is estimated from the surrounding wall texture with detected cracks dilated and excluded, so alignment is driven by surface features independently of the identity under test. This independence is conditional on the masks and the succeeding stage, not guaranteed per pair. Full detector settings (SIFT with CLAHE [36], ratio test, robust estimation) are archived with the code.

B. Registration cascade and validity checks

Textured render generally provides background features, whereas smooth plaster does not. A three-stage cascade is used: masked SIFT; if too few keypoints remain, unmasked SIFT (flagged, crack leakage occurs); finally, photometric ECC [34]. Candidate solutions are tested for plausibility (no mirrored or folded quads; scale within $[1/16, 16]$; no inlier-count criterion). We record the surviving stage per pair, and the inlier RMSE at that stage determines the comparison tolerance. Absence of registration means lack of evidence, not that the cracks differ.

C. Crack projection and geometric comparison

Accepted registrations project instances from I_a into I_b and compare them using the symmetric mean Chamfer distance [37] as in (1); this allows small misalignments by using IoU on narrow cracks. Let $aandb$ be crack point sets and $d(p,b) = \min_{q\in b}\|p-q\|_2$:

$$D(a,b) = \frac{1}{2}\left[\frac{1}{|a|}\sum_{p\in a} d(p,b) + \frac{1}{|b|}\sum_{q\in b} d(q,a)\right]. \quad (1)$$

We reject candidates that are far away. Chamfer coverage (fraction within the per-pair RMSE, distance clamped at $[2, 16]$ px, default 6 px) is the scale-free equivalent.

D. Correspondence decisions and limits of spatial evidence

For each pair of photographs we carry out one-to-one matching using the Hungarian method [38] at the validation threshold; unregistered pairs score $-\infty$ (Sec. IV). Alignment of an image with an empty wall acts as evidence of absence; missed detections, out-of-frame areas, and registration failures remain unresolved. The reference targets show no change in damage; performance is determined by the surface and by the origin of the label.

VI. OFF-THE-SHELF APPEARANCE BASELINES VS FULL-PHOTOGRAPH REFERENCE

A. Setup

Ten matchers use crack-centred boxes with a common mask: SIFT/ORB with RANSAC inliers, SuperGlue/LoFTR with indoor weights, five embeddings and OSNet with cosine on

TABLE I

MAIN RESULT, TEST (11 WALLS, n_Q=280 KNOWN, 258 UNKNOWN). POINT ESTIMATES ONLY; EFFECTIVE SAMPLE IS WALLS. METRICS: R@1 RANK-1 ACCURACY; MAP MEAN AVERAGE PRECISION; DIR DETECTION-AND-IDENTIFICATION RATE AT FAR 0.1 (FRACTION OF KNOWN QUERIES CORRECTLY IDENTIFIED WHEN FALSE ALARM RATE $\leq$0.1); F_1^* ORACLE-THRESHOLD F_1; ΔF_1^* EXCESS OVER CHANCE (0.305); $F_1^{@v}$ VALIDATION-THRESHOLD F_1; REG. FRACTION OF TEST PAIRS THAT REGISTER. BOLD: BEST CROP PER COLUMN. GEOMETRY SCORES 48% OF VALID PAIRS AND RANKS THE REST LAST, SO DIR 0.768 VS 0.429 USE DIFFERENT CANDIDATE POOLS.

Method	Family	R@1	mAP	DIR @FAR = 0.1	F_1^*	ΔF_1^*	$F_1^{@v}$	reg.
Chance (random ranking)								
Chance	reference	0.336	0.385	0.038	0.305	0.000	0.305	1.00
Reference level (same 0.48 pool as geometry)								
REG-ONLY	geometry	0.515	0.533	0.000	0.544	0.000	0.544	0.48
Crop scope								
ORB [15]	keypoint	0.352	0.364	0.018	0.305	0.000	0.305	1.00
SIFT [14]	keypoint	0.432	0.417	0.036	0.313	0.008	0.305	1.00
SuperGlue [16]	learned	0.562	0.455	**0.429**	0.353	0.048	0.305	1.00
LoFTR [17]	learned	0.549	0.440	0.336	**0.359**	**0.054**	**0.351**	1.00
YOLOv8 backbone [22]	embedding	0.561	0.453	0.157	0.357	0.052	0.308	1.00
ViT-B/16 [18]	embedding	**0.654**	0.511	0.286	0.331	0.026	0.317	1.00
DeiT-S [19]	embedding	0.621	0.497	0.250	0.305	0.000	0.293	1.00
CLIP ViT-B/32 [20]	embedding	0.604	0.471	0.143	0.307	0.002	0.306	1.00
DINOv2 [21]	embedding	0.607	0.484	0.307	0.305	0.000	0.297	1.00
OSNet [3]	re-ID	**0.654**	**0.520**	0.168	0.309	0.004	0.309	1.00
Full-photograph scope								
REG+CHAMFER (reference)	geometry	0.952	0.886	0.768	0.656	+0.112	0.582	0.48

L2 crops, all without CRACKID tuning (YOLOv8: layer-9 `yolov8n.pt`; sizes/checkpoints in code). Context expands the box $1\times$ per side; erased views inpaint via Telea after 5 px dilation (artefacts possible; sibling/neighbour handling unstated). Only OSNet/CLIP tested, no crack-only/wall-only views. REG-ONLY scores a constant on registered pairs. Appearance sees tight crops, geometry full photographs. Missing: full-photo descriptors (DINOv2, NetVLAD [24]), LightGlue [23], matcher-fed Chamfer (see repo for timings; sessions not load-controlled, no ranking claimed).

B. Result: ranking spreads, decisions stay near the constant baseline

Ranking spreads while thresholded decisions stay near the constant predictor (Table I). Rank-1 0.352–0.654; mAP to 0.520; crop DIR to 0.429. Oracle F_1^* 0.305–0.359; validated $F_1^{@v}$ 0.293–0.351 (excess -0.012 to $+0.012$ except LoFTR $+0.046$; ORB/SIFT/SuperGlue accept everything at 0.305). AUROC/AUPRC, DIR–FAR curves, thresholds and wall-clustered confidence intervals need archived matrices; leave-one-wall-out over 17 walls should replace the 6/11 split. Part of the drop is prevalence shift (0.292 to 0.180), not only appearance. With 19 identities on 11 walls, we claim no fine-grained ranking or separability from chance.

C. Is it the crack, or the wall behind it?

For OSNet, enlarging the crop raises DIR from 0.168 to 0.404 at FAR 0.1; erasing the crack retains DIR at 0.393 (Rank-1: 0.711 vs 0.775), suggesting that location-specific context carries much of the signal. For CLIP, DIR rises from 0.143 to 0.311 with context and 0.286 after erasure, showing a similar pattern. Deltas are merely point estimates without confidence intervals. The $1\times$ box mostly adds neighbouring cracks, possibly sibling fragments (45% merged), so the signal source is context around cracks rather than wall texture alone. Crack-only and wall-only views on additional backbones with confidence intervals would sharpen this interpretation. F_1^* stays 0.307–0.361.

VII. RESULTS AND DISCUSSION

A. Main comparison and the open-set gap

For the registered cases, which make up 48% (all 5,722 positive examples), geometry is used to compare masks whereas the crops evaluate all the pairs from different pools. When registration is used alone (with REG-ONLY set to a constant on registered cases and to $-\infty$ otherwise), the results are Rank-1 0.515, mAP 0.533, and DIR 0.000 at FAR 0.1; when mask comparison is added, the values improve by +0.437 and +0.353 and DIR reaches 0.768 (Table I). The DIR scores are 0.768 for geometry, 0.000 for REG-ONLY (tied scores), 0.429 for SuperGlue, and 0.404 for OSNet+ctx (chance 0.038); there was no wall-clustered test and no claim of separability. The validated F_1 excess is +0.038 (0.582 vs 0.544) over LoFTR's +0.046, indicating that the appearance/geometry contrast is overstated. The advantage obtained through gating would disappear if cross-visit testing were carried out; further research is still needed on the gated-appearance and frame-gap baselines.

B. Registration rate, failures, and ablations

Unregistered pairs are ranked last and predicted as negative (Sec. IV). The gate is label-informative: among registered pairs, all positives are retained, but 36.8% of negatives are discarded, increasing prevalence from 0.180 to 0.373 and lifting the F_1 floor from 0.305 to 0.544.

TABLE II
SYNTHETIC REVISIT SCENARIO: HYBRID-SKEL VERSUS OSNET+CTX, WITH THE F_1 SCORE AT THE VALIDATION THRESHOLD; ONE SEED AND FIVE WALLS. HYBRID-SKEL DIFFERS FROM REG+CHAMFER AS IN TABLE I.

Protocol	Method	n_q	R@1	mAP	DIR @FAR$_{0.1}$	$F_1^{@v}$	F_1^a
A (growth)	HYBRID-SKEL	69	0.275	0.401	0.343	0.569	0.646
	OSNet+ctx	69	**0.826**	**0.647**	**0.739**	0.601	0.631
B (appear.)	HYBRID-SKEL	82	0.274	0.403	0.342	0.574	0.640
	OSNet+ctx	82	**0.671**	**0.570**	**0.663**	0.594	0.609

The comparison shows an improvement of +0.112 for Oracle (F_1^* 0.656) but only +0.038 for validated results ($F_1^{@v}$ 0.582). The failures depend on the wall (Sec. III); for adjacent frames, all 280 queries are answered (median gap is 1), which is why a frame baseline is required. We define sharpness as the Laplacian variance at quarter scale. A test-only logistic indicates that sharpness is better than gap (clustered z 5.11 vs -2.78), but the p value overestimates the precision (11 clusters); the sharpness $\geq$ 10 gate is determined post hoc. The scope of the smooth-plaster failure is not quantified. The ablations for registration/ECC/exclusion were carried out on the test set; the geometry settings should be fixed on the validation set before any final comparison.

C. Synthetic revisit probe

From five walls, using one seed, ten source photographs show a small warp either under (A) one-tip growth or (B) blur/noise/contrast with the structure remaining unchanged, and then they are matched with the actual images (p90 tilt 75°; procedure as in the code). HYBRID-SKEL (τ=8.0 px) differs from REG+CHAMFER (Table I). For ranking/open-set, OSNet+ctx outperforms HYBRID-SKEL (R@1 0.826/0.671 vs 0.275/0.274; DIR 0.739/0.663 vs 0.343/0.342; $F_1^{@v}$ similar at 0.601/0.594 vs 0.569/0.574); assignment-F_1^a shows a narrow reversal (0.646/0.640 vs 0.631/0.609; no pure-homography cell). With respect to RQ3, this single-seed probe provides no evidence of real multi-visit robustness, which remains untested (Sec. VIII). Table II compares HYBRID-SKEL and OSNet+ctx.

VIII. LIMITATIONS AND FUTURE WORK

This study presents the first measurable test of re-identifying a crack in a single visit using a geometric reference. The results are derived from a small number of clustered walls, labels provided by a single annotator, and partial registration coverage. Each of the points listed here corresponds to a finding together with a step for verifying it.

1) *Scale to more walls and visits.* Collect data from more walls, different surface types, and additional annotators to improve the confidence intervals and the reliability of the rankings. Most importantly, include a second session using different distances and lighting conditions to establish a genuine cross-visit benchmark.
2) *Add provenance-free annotation.* One annotator reviewed all clicks and handled the unclaimed points; agreement was not assessed, and the geometric advantage depends on provenance. To detect the gap independent of provenance, create a small set of labels without homography and share score matrices per pair for sensitivity to wall-clustered intervals and threshold-free AP/AUROC.
3) *Close the coverage gap.* The baseline covers 48% of the pairs (97 of 301 image pairs; 15,328 of 31,822 valid cells) and is ineffective on smooth plaster. By using surface-adaptive registration such as learned local features, direct photometric alignment, or sparse depth from consumer devices, a more thorough evaluation could be achieved, and separate geometry could be better distinguished from homography-based provenance.
4) *Learn from geometric supervision.* The baseline is an untrained reference and not a system that has been put into operation. The location benefit could be obtained by using pseudo-labels from the registration process or by fine-tuning a foundation model on homography correspondences, without per-pair registration.
5) *Assess the ticket lifecycle.* Define task metrics such as the duplicate rate and the missed-new-defect rate, and also test segmentation, re-identification, and joint ticket assignment in a closed-loop simulation of real inspections.

Together, these steps will turn the within-visit reference into a multi-visit re-identification component that can be used operationally for tickets.

IX. CONCLUSION

We present CRACKID, which consists of 140 photographs from 17 walls, along with a protocol for retrieval, deciding whether defects are the same, and performing open-set identification with rejection.

The main result is negative and structural in character: concerning RQ1, ten zero-shot crop matchers outperform random chance, but when a threshold is used, their performance only just exceeds that of the best constant baseline; concerning RQ2, the wall context has the biggest effect on the appearance of cracks, and a geometric reference assists with open-set identification in only the 48% of the alignable subset that has been conditioned on annotation provenance; as for RQ3, a small synthetic revisit probe reverses the ranking with no sign of multi-visit robustness.

The study focuses on single-visit photographs: the figures are point estimates derived from 19 test identities, grouped by wall and matched using only the labels supplied by one annotator during the visual-association stage. The next crucial stage is to conduct an acquisition session to run a cross-visit benchmark at the ticket level.

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REFERENCES

[1] S. Kulkarni, S. Singh, D. Balakrishnan, S. Sharma, S. Devunuri, and S. C. R. Korlapati, "CrackSeg9k: A collection and benchmark for crack segmentation datasets and frameworks," arXiv:2208.13054 [cs.CV], 2022.

[2] C. Benz and V. Rodehorst, "OmniCrack30k: A benchmark for crack segmentation and the reasonable effectiveness of transfer learning," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2024.

[3] K. Zhou, Y. Yang, A. Cavallaro, and T. Xiang, "Omni-scale feature learning for person re-identification," in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019, pp. 3702–3712.

[4] Y. Qian, J. Barthelemy, E. Karuppiah *et al.*, "Identifying re-identification challenges: Past, current and future trends," *SN Computer Science*, vol. 5, p. 937, 2024.

[5] Q. Zou, Y. Cao, Q. Li, Q. Mao, and S. Wang, "CrackTree: Automatic crack detection from pavement images," *Pattern Recognition Letters*, vol. 33, no. 3, pp. 227–238, 2012.

[6] Y. Liu, J. Yao, X. Lu, R. Xie, and L. Li, "DeepCrack: A deep hierarchical feature learning architecture for crack segmentation," *Neurocomputing*, vol. 338, pp. 139–153, 2019.

[7] M. Eisenbach, R. Stricker, D. Seichter, K. Amende, K. Debes, M. Sesselmann, D. Ebersbach, U. Stoeckert, and H.-M. Gross, "How to get pavement distress detection ready for deep learning? a systematic approach," in *Int. Joint Conf. Neural Networks (IJCNN)*, 2017, pp. 2039–2047.

[8] C. Benz and V. Rodehorst, "CrackStructures and CrackEnsembles: The power of multi-view for 2.5d crack detection," in *Proceedings of the Winter Conference on Applications of Computer Vision (WACV)*, 2025, pp. 5990–5999.

[9] L. Parente, E. Falvo, C. Castagnetti, F. Grassi, F. Mancini, P. Rossi, and A. Capra, "Image-based monitoring of cracks: Effectiveness analysis of an open-source machine learning-assisted procedure," *Journal of Imaging*, vol. 8, no. 2, p. 22, 2022.

[10] S. Harb, P. Achanccaray, M. Maboudi, and M. Gerke, "Multi-temporal crack segmentation in concrete structures using deep learning approaches," *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, vol. X-G-2025, pp. 341–348, 2025.

[11] X. Sun and P. Chang, "Robust perspective correction for real-world crack evolution tracking in image-based structural health monitoring," arXiv:2506.22437 [cs.CV], 2025.

[12] NEC Corporation, "Photographing guide device," US Patent 12,223,639 B2 (priority 2019), 2025.

[13] S. Liao, Z. Mo, Y. Hu, and S. Z. Li, "Open-set person re-identification," arXiv:1408.0872 [cs.CV], 2014.

[14] D. G. Lowe, "Distinctive image features from scale-invariant keypoints," *International Journal of Computer Vision*, vol. 60, no. 2, pp. 91–110, 2004.

[15] E. Rublee, V. Rabaud, K. Konolige, and G. Bradski, "ORB: An efficient alternative to SIFT or SURF," in *IEEE International Conference on Computer Vision (ICCV)*, 2011, pp. 2564–2571.

[16] P.-E. Sarlin, D. DeTone, T. Malisiewicz, and A. Rabinovich, "SuperGlue: Learning feature matching with graph neural networks," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 4938–4947.

[17] J. Sun, Z. Shen, Y. Wang, H. Bao, and X. Zhou, "LoFTR: Detector-free local feature matching with transformers," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 8922–8931.

[18] A. Dosovitskiy, L. Beyer, A. Kolesnikov *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," in *International Conference on Learning Representations (ICLR)*, 2021.

[19] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, and H. Jégou, "Training data-efficient image transformers and distillation through attention," in *International Conference on Machine Learning (ICML)*, 2021, pp. 10 347–10 357.

[20] A. Radford, J. W. Kim, C. Hallacy *et al.*, "Learning transferable visual models from natural language supervision," in *International Conference on Machine Learning (ICML)*, 2021, pp. 8748–8763.

[21] M. Oquab, T. Darcet, T. Moutakanni *et al.*, "DINOv2: Learning robust visual features without supervision," *Transactions on Machine Learning Research*, 2024.

[22] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics YOLOv8 (software)," Available: https://github.com/ultralytics/ultralytics, 2023.

[23] P. Lindenberger, P.-E. Sarlin, and M. Pollefeys, "LightGlue: Local feature matching at light speed," in *IEEE/CVF Int. Conference on Computer Vision (ICCV)*, 2023, pp. 17 627–17 638.

[24] R. Arandjelović, P. Gronat, A. Torii, T. Pajdla, and J. Sivic, "NetVLAD: CNN architecture for weakly supervised place recognition," in *IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 5297–5307.

[25] L. Maiano, A. Montuschi, M. Caserio *et al.*, "A deep-learning-based antifraud system for car-insurance claims," *Expert Systems with Applications*, vol. 231, p. 120644, 2023.

[26] M. J. Hasan, C. K. Nguyen, Y. L. Boo, H. Jahani, and K.-L. Ong, "Vehicle damage detection using artificial intelligence: A systematic literature review," *WIREs Data Mining and Knowledge Discovery*, vol. 15, p. e70027, 2025.

[27] S. A. Pérez-Zarate, D. Corzo-García, J. L. Pro-Martín *et al.*, "Automated car damage assessment using computer vision: Insurance company use case," *Applied Sciences*, vol. 14, no. 20, p. 9560, 2024.

[28] D. Mallios, L. Xiaofei, N. McLaughlin, J. M. Del Rincon, C. Galbraith, and R. Garland, "Vehicle damage severity estimation for insurance operations using in-the-wild mobile images," *IEEE Access*, vol. 11, pp. 78 644–78 655, 2023.

[29] K. Sakurada and T. Okatani, "Change detection from a street image pair using CNN features and superpixel segmentation," in *British Machine Vision Conference (BMVC)*, 2015.

[30] P. F. Alcantarilla, S. Stent, G. Ros, R. Arroyo, and R. Gherardi, "Street-view change detection with deconvolutional networks," *Autonomous Robots*, vol. 42, no. 7, pp. 1301–1322, 2018.

[31] R. Hartley and A. Zisserman, *Multiple View Geometry in Computer Vision*, 2nd ed. Cambridge University Press, 2004.

[32] M. A. Fischler and R. C. Bolles, "Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography," *Communications of the ACM*, vol. 24, no. 6, pp. 381–395, 1981, reprinted in Readings in Computer Vision, Morgan Kaufmann, 1987.

[33] D. Barath, J. Matas, and J. Noskova, "MAGSAC: Marginalizing sample consensus," in *IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 10 197–10 205.

[34] G. D. Evangelidis and E. Z. Psarakis, "Parametric image alignment using enhanced correlation coefficient maximization," *IEEE Trans. Pattern Analysis and Machine Intelligence*, vol. 30, no. 10, pp. 1858–1865, 2008.

[35] D. Ferramenta, "CrackID," Mendeley Data, Version 1, Available: https://doi.org/10.17632/t749fc4x44.1, 2026.

[36] K. Zuiderveld, "Contrast limited adaptive histogram equalization," in *Graphics Gems IV*. Academic Press, 1994, pp. 474–485.

[37] H. G. Barrow, J. M. Tenenbaum, R. C. Bolles, and H. C. Wolf, "Parametric correspondence and chamfer matching: Two new techniques for image matching," in *Int. Joint Conf. Artificial Intelligence (IJCAI)*, 1977, pp. 659–663.

[38] H. W. Kuhn, "The Hungarian method for the assignment problem," *Naval Research Logistics Quarterly*, vol. 2, no. 1-2, pp. 83–97, 1955.